

**23rd NATIONAL CERTIFICATION EXAMINATION
FOR
ENERGY MANAGERS & ENERGY AUDITORS - MARCH, 2023**

PAPER – 4 : ENERGY PERFORMANCE ASSESSMENT FOR EQUIPMENT AND UTILITY SYSTEMS

Date : 26.03.2023 Timings : 14:00-16:00 HRS Duration : 2 HRS Max. Marks : 100

Section - I: BRIEF QUESTIONS

Marks: 10 x 1 = 10

- (i) Answer all **Ten** questions
(ii) Each question carries **One** mark

1	If an electric motor is operated at the rated frequency and voltage without any shaft load, then the Power Factor will be high.	True/False	False
2	For the same head and flow, a pump operating with water will consume less power than operating with kerosene.	True/False	False
3	The higher the TTD (Terminal Temperature Difference) and the DCA (Drain Cooler Approach), the more efficient is the Rankine Cycle.	True/False	False
4	For a centralized air conditioning system with an air-cooled condenser the kW/TR will be lesser than the one with water cooled condenser.	True/False	False
5	The direct method of efficiency calculation of boilers does not require steam generation pressure.	True/False	False
6	A slight negative pressure is maintained in the heating zone of the furnace to maintain uniform heating of the stock.	True/False	False
7	In a thermal power plant, the boiler efficiency affects the turbine heat rate.	True/False	False
8	Higher the slip, higher will be the loading of motor.	True/False	True
9	A roof exhaust fan in an industrial shed belong to the category of axial flow fans.	True/False	True
10	A screw compressor designed for 7 bar if operated at 5 bar, then its FAD will decrease.	True/False	False

..... **End of Section - I**

Section - II: SHORT NUMERICAL QUESTIONS

Marks: 2 x 5 = 10

L1	The flue gas analysis of an Industrial Boiler revealed CO : 5000 PPM and CO ₂ :14%. The Gross Calorific Value of the fuel is 4000 kCal/kg. The fuel has 36 kg of carbon per 100 kg of fuel. Estimate the percentage loss due incomplete combustion of C to CO.
Ans	The heat Loss due to Partial Combustion = $[(\%CO \times C)/(\%CO + \%CO_2)] \times 5654$ CO = 5000 PPM = $5000 \times 100 / 1000000 = 0.5$ CO₂ = 14 = $[(0.5 \times 0.36) / (0.5 + 14)] \times 5654 = 70.19 \text{ kCal}$ GCV of Fuel = 4000 kCal % loss due to incomplete combustion = $70.19 / 4000 \times 100 = 1.75\%$
L2	The operating data of Vapor Absorption Chiller is given below: Chilled Water Flow : 0.18 m³/sec Specific Heat of Chilled Water : 4.18 kJ/kg/K Density of water : 1000 kg/m³ Chilled Water Inlet Temperature : 9.0 °C Chilled Water Outlet Temperature : 7.0 °C Enthalpy of Steam : 2750 kJ/kg Steam Flow Rate : 1 kg/sec Condensate Temperature : 100 °C Calculate the following: A) Refrigeration effect in TR B) Thermal input energy in kJ/s C) Coefficient of performance (COP)
Ans	A) Refrigeration Effect in TR Refrigeration Effect = $M_c \times \rho \times C_p \times (t_{cin} - t_{cout}) \text{ kJ/s}$ = $0.18 \times 1000 \times 4.18 \times (9 - 7)$ = 1504.8 kJ/s = $1504.8 / 3.516 \text{ TR}$ = 428 TR B) Thermal Input Energy Input Energy = $M_{st} \times (h_{st} - h_{cond})$ = $1 \times (2750 - 418)$ = 2332 kJ/s C) Coefficient of Performance (COP) = Refrigeration Effect/Thermal Input Energy = $1504.8 / 2332 = 0.65$

..... **End of Section - II**

Section - III: LONG NUMERICAL QUESTIONS

Marks: 4 x 20 = 80

(i) Answer all the **Four** questions

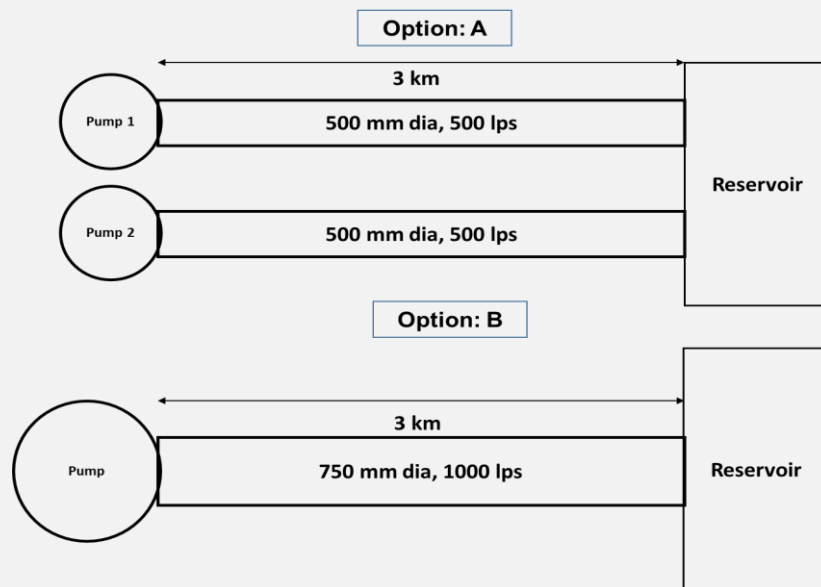
(ii) Each question carries **Twenty** marks

N-1 A)

A factory having a water requirement of 1000 lit/sec is planning to draw water from river basin located 3 km away from the plant. The following two options are evaluated:

Option A: Two separate pumps connected to separate dedicated discharge line of 500 mm dia with 3 km length and each pump delivering 500 l/s. The efficiency of the pump is 80% and motor efficiency is 93%.

Option B: Single pump connected to a discharge line of 750 mm diameter with 3 km length and delivering 1000 l/s. The efficiency of the pump is 80% and motor efficiency is 95%.



The other data's are follows:

System static head	: 12 m
Pipe friction factor is	: 0.006
Cost of energy	: ₹10 /kWh
Annual operating hours	: 8000

As an energy auditor which option you will recommend considering the annual energy cost.

[15 Marks]

B)

A cooling tower connected to the condenser of a refrigeration system is operating with a heat load of 280×10^3 kCal/hr. The hot water inlet temperature to the cooling tower and approach are 33°C and 3°C respectively. The wet bulb temperature is 25°C. The condenser pump is operating at a total head of 25m. The combined efficiency of pump and motor is 54 %. Calculate the power drawn by the condenser pump.

[5 Marks]

Ans	A)			
	Parameter	Two separate pumps connected to separate dedicated discharge line of 500 mm dia with 3 km length and each pump delivering 500 l/s. The efficiency of the pump is 80% and motor efficiency is 93%.	Single pump connected to a discharge line of 750 mm diameter with 3 km length and delivering 1000 l/s. The efficiency of the pump is 80% and motor efficiency is 95%.	Units
	Discharge flow (Q)	500	1000	lps
		0.5	1	m ³ /s
	Diameter of pipe	0.5	0.75	m
	Area	$=3.141 \times 0.5 \times 0.5 / 4$ = 0.196	$=3.141 \times 0.75 \times 0.75 / 4$ = 0.442	m ²
	Velocity	$=0.5 / 0.196$ = 2.55	$= 1 / 0.442$ = 2.26	m/s
	Length of Pipe	3000	3000	M
	Pipe friction factor	0.006	0.006	
	Friction resistance	$= 4flv^2 / 2gD$ $= 4 \times 0.006 \times 3000 \times 2.55^2 / (2 \times 9.81 \times 0.5)$ = 47.61	$= 4flv^2 / 2gD$ $= 4 \times 0.006 \times 3000 \times 2.26^2 / (2 \times 9.81 \times 0.75)$ = 25.08	M
	Static Head	12	12	M
	Total head	$= 47.61 + 12$ = 59.61	$= 25.08 + 12$ = 37.08	M
	Combined Efficiency (Pump & Motor)	$= 0.8 \times 0.93$ = 0.744	$= 0.8 \times 0.95$ = 0.76	
	Hydraulic power, P _h	$= Q \times H \times \rho \times g / 1000$ $= 0.5 \times 59.61 \times 1000 \times 9.81 / 1000$ = 292.39	$= Q \times H \times \rho \times g / 1000$ $= 1 \times 37.08 \times 1000 \times 9.81 / 1000$ = 363.75	kW
	Motor input power	$= \text{Hydraulic Power} / \text{Combined Eff}$ $= 292.39 / 0.744 = 393.00$	$= \text{Hydraulic Power} / \text{Combined Eff}$ $= 363.75 / 0.76 = 478.61$	kW
	No. of pumps	2	1	nos.
	Total power drawn	$= \text{Motor input power} \times \text{No of pumps}$ $= 393 \times 2 = 786.00$	$= \text{Motor input power} \times \text{No of pumps}$ $= 478.61 \times 1 = 478.61$	kw
	No. of hours of operation	8000	8000	hours

Total units consumed	= Total power drawn x no of opr hrs = 786 x 8000 = 62,88,000	= Total power drawn x no of opr hrs = 478.61 x 8000 = 38,28,880	kWh
Energy Cost	10	10	₹/kWh
Annual Energy cost	= Total units consumed x Cost ₹ 6,28,80,000	= Total units consumed x Cost ₹ 3,82,88,800	₹/year
Cost Reduction	₹ 2,45,91,200		₹

Option B is recommended considering the annual energy cost.

B)

Cooling Tower Heat Rejection	280000	kCal/hr
Cold water temperature of basin	= 25+3 = 28	°C
Range of Cooling tower	= (33- 28) = 5	°C
Flow Rate of Cooling Water	= (280000/5)/1000 = 56.0	m ³ / hr
Head of Pump	25	m
Combined Efficiency	54	%
Hydraulic Power	= $Q \cdot H \cdot \rho \cdot g / 1000$ = ((56/3600)* 25*1000*9.81)/1000 = 3.82	kW
Power Drawn by the Pump	= Hydraulic Power / Combined Eff = 3.82/54*100 = 7.1	kW

N-2 An oil refinery has captive power plant with pet coke fired boiler. The following are the data collected to assess the boiler performance.

Pet coke elementary analysis	
Carbon	88.8
Hydrogen	3.6
Nitrogen	1.2
Oxygen	1.4
Sulphur	3.6
Moisture	1.4

GCV of Petcoke : 8430 kCal/kg
Stack O₂ : 6.0 mol%
Stack Temperature : 250 °C
Heat loss due to radiation & convection : 1.0 %
Loss due to unburnt in fly ash & bottom ash : 0.5%
Specific heat of flue gas : 0.29 kCal/kg °C
Specific heat of water vapour : 0.45 kCal/kg °C
Ambient Temperature : 30 °C

	Humidity in ambient air : 0.0204 kg/kg dry air Steam generation at 110 bar g & 520 °C Steam enthalpy at generation pressure and temperature : 816 kCal/kg Feed water temperature : 200 °C Steam drum pressure : 115 bar g A) Calculate Boiler efficiency using indirect method. [12] B) Calculate evaporation ratio. [3] C) Calculate the quantity (kg) flash steam generated per kg of blow down if: [5] - Saturated liquid enthalpy at steam drum pressure: 352 kCal/kg - Total enthalpy flash steam at 2 bar (g) pressure: 646 kCal/kg - Latent heat of evaporation at 2 bar (g) pressure: 526 kCal/kg																																		
Ans	<table border="1"> <tr> <td>A)</td><td>Theoretical Air required</td><td> $= \{(11.6 \times 0.888) + [34.8 \times (0.036 - 0.014/8)] + 4.35 \times 0.036\}$ $= 11.65$ </td><td>kg air / kg of petcoke</td></tr> <tr> <td></td><td>% Excess Air supplied</td><td> $= 100 \times 6.0 / (21 - 6.0)$ $= 40$ </td><td>%</td></tr> <tr> <td></td><td>Actual Air supplied</td><td> $= (1 + 40/100) \times 11.65$ $= 16.31$ </td><td>kg air / kg of petcoke</td></tr> <tr> <td></td><td>Mass of dry flue gas</td><td> $= \{(0.888 \times 44/12) + 0.012 + (16.31 \times 0.77) + (16.31 - 11.65) \times 0.23 + (0.036 \times 64/32)\}$ $= 16.97$ Or $= (16.31 + 1) - (0.014 + (9 \times 0.036))$ $= 16.97$ </td><td>kg air / kg of petcoke</td></tr> <tr> <td></td><td>Stack losses, L1</td><td> $= 16.97 \times 0.29 \times (250 - 30) / 8430 \times 100$ $= 12.84$ </td><td>%</td></tr> <tr> <td></td><td>Loss due to formation of water vapor from H₂ in fuel, L2</td><td> $= 9 \times 0.036 \times \{584 + 0.45 \times (250 - 30)\} / 8430 \times 100$ $= 2.63$ </td><td>%</td></tr> <tr> <td></td><td>Loss due to moisture in fuel, L3</td><td> $= 0.014 \times \{584 + 0.45 \times (250 - 30)\} / 8430 \times 100$ $= 0.11$ </td><td>%</td></tr> <tr> <td></td><td>Loss due to moisture in Air, L4</td><td> $= 16.31 \times 0.0204 \times 0.45 \times (250 - 30) / 8430 \times 100$ </td><td>%</td></tr> </table>			A)	Theoretical Air required	$= \{(11.6 \times 0.888) + [34.8 \times (0.036 - 0.014/8)] + 4.35 \times 0.036\}$ $= 11.65$	kg air / kg of petcoke		% Excess Air supplied	$= 100 \times 6.0 / (21 - 6.0)$ $= 40$	%		Actual Air supplied	$= (1 + 40/100) \times 11.65$ $= 16.31$	kg air / kg of petcoke		Mass of dry flue gas	$= \{(0.888 \times 44/12) + 0.012 + (16.31 \times 0.77) + (16.31 - 11.65) \times 0.23 + (0.036 \times 64/32)\}$ $= 16.97$ Or $= (16.31 + 1) - (0.014 + (9 \times 0.036))$ $= 16.97$	kg air / kg of petcoke		Stack losses, L1	$= 16.97 \times 0.29 \times (250 - 30) / 8430 \times 100$ $= 12.84$	%		Loss due to formation of water vapor from H ₂ in fuel, L2	$= 9 \times 0.036 \times \{584 + 0.45 \times (250 - 30)\} / 8430 \times 100$ $= 2.63$	%		Loss due to moisture in fuel, L3	$= 0.014 \times \{584 + 0.45 \times (250 - 30)\} / 8430 \times 100$ $= 0.11$	%		Loss due to moisture in Air, L4	$= 16.31 \times 0.0204 \times 0.45 \times (250 - 30) / 8430 \times 100$	%
A)	Theoretical Air required	$= \{(11.6 \times 0.888) + [34.8 \times (0.036 - 0.014/8)] + 4.35 \times 0.036\}$ $= 11.65$	kg air / kg of petcoke																																
	% Excess Air supplied	$= 100 \times 6.0 / (21 - 6.0)$ $= 40$	%																																
	Actual Air supplied	$= (1 + 40/100) \times 11.65$ $= 16.31$	kg air / kg of petcoke																																
	Mass of dry flue gas	$= \{(0.888 \times 44/12) + 0.012 + (16.31 \times 0.77) + (16.31 - 11.65) \times 0.23 + (0.036 \times 64/32)\}$ $= 16.97$ Or $= (16.31 + 1) - (0.014 + (9 \times 0.036))$ $= 16.97$	kg air / kg of petcoke																																
	Stack losses, L1	$= 16.97 \times 0.29 \times (250 - 30) / 8430 \times 100$ $= 12.84$	%																																
	Loss due to formation of water vapor from H ₂ in fuel, L2	$= 9 \times 0.036 \times \{584 + 0.45 \times (250 - 30)\} / 8430 \times 100$ $= 2.63$	%																																
	Loss due to moisture in fuel, L3	$= 0.014 \times \{584 + 0.45 \times (250 - 30)\} / 8430 \times 100$ $= 0.11$	%																																
	Loss due to moisture in Air, L4	$= 16.31 \times 0.0204 \times 0.45 \times (250 - 30) / 8430 \times 100$	%																																

		= 0.39	
	Loss due to radiation and convection	= 1.0	%
	Loss due to unburnt in fly ash & bottom ash	= 0.5	%
	Efficiency of boiler using indirect method	= (100-12.84-2.63-0.11-0.39-1-0.5) = 82.53	%
B)	Heat required per unit of steam generation	= (816-200) = 616.0	kcal/kg of Steam
	Heat supplied by fuel for steam generation	= (82.5/100)x8430 = 6957.28	kcal/kg of petcoke
C)	Evaporation Ratio Quantity (kg) flash steam generated per kg of blow down	= 6957.28/616 = 11.29 S1 = 352 Kcal/kg S2 = 646 – 526 = 120 kCal/kg L2 = 526 Kcal kg % Flash Steam quantity from blowdown: (352-120)/526 = 44.1% Quantity of flash steam per kg of blow down: 0.44 kg/ kg of blow down.	MT Steam/MT of petcoke

N-3	In an engineering company, the compressed air is required for pneumatic equipment's and other processes. To meet the requirement, one reciprocating compressor of 500 CFM is installed. To assess the performance of compressor, Free Air Delivery test was carried out. The test and other data are given below. Receiver Capacity : 10 m³ Interconnecting pipe : 1 m³ Initial Pressure in Receiver : 1.0 kg/cm² a Atmospheric Pressure : 1.0 kg/cm² a Final Pressure : 8.25 kg/cm² a Time Taken to Fill Receiver : 6 Minute Inlet air temperature (T_{in}) : 30 °C Air temperature in the receiver (T_{out}) : 40 °C Motor RPM : 1400 RPM Motor Efficiency : 93% Motor Pulley Diameter : 300 mm Compressor RPM : 700 RPM Compressor Pulley Diameter : 600 mm Average Loading Time : 65% Average Unloading Time : 35% Power Consumption during loading : 82 kW Power Consumption during unloading : 21 kW		
------------	---	--	--

	Cost of Energy : 10.00 Rs / kWh Answer the followings: A) Estimate the Free Air Deliver of Compressor? [5 Marks] B) Evaluate the operating energy cost per day (for 24 hours of operation) [2 Marks] C) Calculate the isothermal power and isothermal efficiency of the compressor. [5 Marks] D) To reduce the energy loss due to unloading, the maintenance team has decided to reduce the speed of the compressor by reducing the motor pulley size. Evaluate the speed of compressor required for 10 min unloading time and 50 min of loading time and accordingly evaluate the diameter of the pulley of the motor. [5 Marks] E) Estimate the hourly energy consumption, energy saving after replacement of the pulley and payback period. The cost of pulley and belts is Rs. 1.2 Lakhs. The operating hours of the compressor is 8000 in a year. Consider power consumption during loading is 64 kW and power consumption during unloading is 18 kW after motor pulley change. [3 Marks]
Ans	A) Free Air Delivery (with temperature correction) $Q = \frac{P_2 - P_1}{P_0} \times \frac{V}{T} \times \frac{(273 + t_1)}{(273 + t_2)}$ $= 12.867 \text{ m}^3/\text{min}$ $= 12.867 \times 35.31 = \mathbf{454.33 \text{ CFM}}$ B) To evaluate the operating energy cost per day (for 24 hours of operation) Reciprocating compressor rated capacity = 500 CFM Percentage with respect to rated capacity = $(500 - 454.33 / 500) \times 100$ $= 9.1 \%$ Loading energy (65% Loading) = Average loading time x Power consumption during loading $= 65 \% \times 82 \text{ kW} = 53.3 \text{ kWh}$ Unloading energy (65% Loading) = Average unloading time x Power consumption during unloading $= 35 \% \times 21 \text{ kW} = 7.35 \text{ kWh}$ Hourly energy consumption = $53.3 + 7.35 = 60.65 \text{ kWh}$ Energy cost per day = $60.65 \times 10 \times 24 = \mathbf{Rs.14556/-}$ C) Calculate the isothermal power and isothermal efficiency of the compressor. To calculate Isothermal Power = $P_1 \times Q_1 \times \log_e r / 36.7$ r = Pressure ratio, P_2/P_1 P_1 = Absolute intake pressure, kg/cm² P_2 = Absolute delivery pressure, kg/cm² Q_1 = Free air delivered, m³/hr $= 1 \times 13.31 \times 60 \times (\ln (8.25/1)/36.7)$ $= \mathbf{44.39 \text{ kW}}$

	To calculate Isothermal efficiency = $\frac{\text{Isothermal Power}}{\text{Actual measured input power}}$ $= 44.39 / (82 \times 0.93) = \mathbf{58.21 \%}$ D) Hourly air delivery = 454.3 CFM x 65%, Average loading time = 295.3 Cu.ft/hr Loading Time Existing (T1) = 65% New Loading Time after pulley change(T2) = 83.3% Existing RPM of Compressor (N1) = 700 RPM New RPM of Compressor after Pulley Change (N2) = $(700 \times 65\%) / 83.3\% = \mathbf{546 \text{ RPM}}$ Existing Motor Pulley Size = 300 MM New Motor Pulley Diameter = $(300 \times 546) / 700 = \mathbf{234 \text{ mm}}$ E) New Loading Power = 64 kW (given) Unloading Power = 18 kW (given) Hourly Energy Consumption after pulley Change = $(64 \times 50/60) + (18 \times 10/60) = 56.33 \text{ kWh}$ Reduction in Power Consumption = $60.65 - 56.33 = 4.32 \text{ kWh}$ Annual Energy reduction = $4.32 \times 8000 = 34533.33 \text{ kWh}$ Annual Cost Reduction = $34533.33 \times 10 = \mathbf{Rs.3,45,333/-}$ Cost of Pulley & Belt Change = 120000 Rs. Simple Payback Period = $120000 / 345333 = \mathbf{4.17 \text{ Months}}$																								
	Answer any ONE of the following among four questions given below:																								
N4 A	The following is the data collected from a 500 MW Turbine unit by an Energy Auditor. <table border="1"> <tbody> <tr> <td>Main Steam Flow (TPH)</td><td>1561</td></tr> <tr> <td>Hot reheat Flow (TPH)</td><td>1413</td></tr> <tr> <td>Main steam pressure (kg/cm²)/ Temperature (°C)</td><td>166/529</td></tr> <tr> <td>Cold reheat (CRH) pressure (kg/cm²)/ Temperature (°C)</td><td>44.3/341</td></tr> <tr> <td>Hot reheat pressure (HRH) (kg/cm²)/ Temperature (°C)</td><td>42.4/540</td></tr> <tr> <td>Feed water temperature (°C)</td><td>246</td></tr> <tr> <td>Main steam enthalpy (kCal/kg°C)</td><td>806.47</td></tr> <tr> <td>Feed water enthalpy (kCal/kg°C)</td><td>246</td></tr> <tr> <td>CRH Enthalpy (kCal/kg°C)</td><td>730.71</td></tr> <tr> <td>HRH Enthalpy (kCal/kg°C)</td><td>844.27</td></tr> <tr> <td>Generator output (MW)</td><td>501.7</td></tr> <tr> <td>Boiler Efficiency (%)</td><td>88%</td></tr> </tbody> </table>	Main Steam Flow (TPH)	1561	Hot reheat Flow (TPH)	1413	Main steam pressure (kg/cm ²)/ Temperature (°C)	166/529	Cold reheat (CRH) pressure (kg/cm ²)/ Temperature (°C)	44.3/341	Hot reheat pressure (HRH) (kg/cm ²)/ Temperature (°C)	42.4/540	Feed water temperature (°C)	246	Main steam enthalpy (kCal/kg°C)	806.47	Feed water enthalpy (kCal/kg°C)	246	CRH Enthalpy (kCal/kg°C)	730.71	HRH Enthalpy (kCal/kg°C)	844.27	Generator output (MW)	501.7	Boiler Efficiency (%)	88%
Main Steam Flow (TPH)	1561																								
Hot reheat Flow (TPH)	1413																								
Main steam pressure (kg/cm ²)/ Temperature (°C)	166/529																								
Cold reheat (CRH) pressure (kg/cm ²)/ Temperature (°C)	44.3/341																								
Hot reheat pressure (HRH) (kg/cm ²)/ Temperature (°C)	42.4/540																								
Feed water temperature (°C)	246																								
Main steam enthalpy (kCal/kg°C)	806.47																								
Feed water enthalpy (kCal/kg°C)	246																								
CRH Enthalpy (kCal/kg°C)	730.71																								
HRH Enthalpy (kCal/kg°C)	844.27																								
Generator output (MW)	501.7																								
Boiler Efficiency (%)	88%																								

Heater Reference	Steam			Feed water in		Feed water out		Design Values	
	Temp p (°C)	Saturati on Temp (°C)	Drain Temp (°C)	Temp (°C)	Pressure (kg/cm ²)	Temp (°C)	Pressur e (kg/cm ²)	TT D (°C)	DCA (°C)
LP Heater 1	73	70	64.2	47.2	13.7	63.6	12.6	2.88	4.8
LP Heater 2	140	111.23	70.4	-	-	105	11.5	2.95	4.95
LP Heater 3	209	132.9	110	-	-	130	10.4	2.95	4.95
HP Heater 5	416	207.33	171	170	202	210	199	0	5
HP Heater 6	335	254.94	212	-	-	255	197	0.1	5

Calculate the following:

- Turbine cycle heat rate and unit gross heat rate. **(6 Marks)**
 - Net heat rate and efficiency of the power plant if the auxiliary power consumption is 6%. **(2 Marks)**
 - Loss/gain in the turbine heat rate because of the deviation of the TTD and DCA of the LP/HP Heater systems from the design values. **(12 Marks)**
- Consider,
- for every 0.56°C increase or decrease of TTD from the design value, Heat Rate will increase or decrease by 0.014%.
 - For every 0.56°C increase or decrease of DCA from the design value, Heat Rate will increase or decrease by 0.005%.

Ans

Turbine heat rate

Turbine heat rate is defined as the amount of heat input to the turbine in kCal for generating one unit of electricity.

$$\text{Turbine heat rate, kCal/kWh} = \frac{Q_1 \times (H_1 - h_2) + Q_2 \times (H_3 - H_2)}{\text{Gross Generator Output}}$$

Q₁ - Main steam flow, kg/hr

H₁ - Main steam enthalpy, kCal/kg

h₂ - Feed water enthalpy, kCal/kg

Q₂ - Reheat steam flow, kg/hr

H₃ - Hot reheat enthalpy, kCal/kg

H₂ - Cold reheat enthalpy, kCal/kg

Unit heat rate

$$\text{Unit heat rate, kCal/kWh} = \frac{\text{Turbine heat rate, kCal/kWh}}{\text{Boiler efficiency}}$$

a.) Turbine Heat Rate:

$$= \{(1561000 \times (806.47 - 246)) + (1413000 \times (844.27 - 730.71))\} / (501700) \\ = 2063.72 \text{ kCal/kWh}$$

$$\text{Unit Gross Heat Rate:} = 2063.72 / 88\% \\ = 2345.131 \text{ kCal/kWh}$$

$$\text{b.) Net Heat Rate} = 2345.131 / (1 - 0.06) = 2494.82 \text{ kCal/kWh} \\ \text{Power plant efficiency} = 860 / 2494.82 \times 100 = 34.5\%$$

c.) From the above data, the following heater data can be inferred

Heater Ref.	Feed water In-let Temp °C	Feed water Out-let Temp °C	Steam Inlet temp °C	Inlet steam Saturation Temp °C	Drain temp °C
LP Heater-1	47.2	63.6	73	70	64.2
LP Heater-2	63.6	105	140	111.23	70.4
LP Heater-3	105	130	209	132.9	110
HP Heater-5	170	210	416	207.33	171
HP Heater-6	210	255	335	254.94	212

So

TTD (Terminal Temperature Difference) = Inlet Steam Saturation Temp °C – Feed Outlet Temp °C

DCA (Drain Cooler Approach Temperature °C) = Drain temperature °C – Feed Water Inlet Temperature °C

Heater Ref.	TTD °C (Design)	DCA °C (Design)	TTD °C (Calculated)	DCA °C (Calculated)
LP Heater -1	2.88	4.8	70-63.6 = 6.4	64.2-47.2 = 17
LP Heater-2	2.95	4.95	6.23	6.8
LP Heater -3	2.95	4.95	2.9	5
HP Heater-5	0	5	-2.67	1
HP Heater -6	0.1	5	-0.06	2

Difference Between design values and Operating values of TTD and DCA of Heaters.

Heater Ref.	TTD _{Operating} – TTD _{Design}	DCA _{Operating} – DCA _{Design}
LP Heater -1	3.52	12.2
LP Heater-2	3.28	1.85
LP Heater -3	-0.05	0.05
HP Heater-5	-2.67	-4
HP Heater -6	-0.16	-3
Total Difference	3.92	7.1

Change in Heat rate because of deviation in TTD =

(Net Change in TTD for All heaters X 0.014%/0.56°C)

Since given, for every 0.56°C change in TTD HR will increase by 0.014%

Increase in HR because of TTD deviation = 3.92 X 0.014 / 0.56 = 0.098 %

Change in Heat rate because of deviation in DCA =

(Net Change in DCA for All heaters X 0.005%/0.56°C)

Since given, for every 0.56°C change in TTD HR will increase by 0.005%

So Increase in HR because of DCA deviation = 7.1 X 0.005 / 0.56 = 0.063 %

Total % Increase in Turbine HR because of deviation in operation of TTD and DCA of Heaters from Design Values

= 0.161 %.

= 0.161% X 2063.72 = 3.32 kCal/kWh

Or

N4 B A cement company manufactures ordinary clinker and low heat clinker. The chemical composition of

the cement clinkers (loss free basis) are given in the following table:

Constituent	Percentage of constituent	
	Ordinary Clinker	Low Heat Clinker
SiO ₂	20	44
Fe ₂ O ₃	7	15
Al ₂ O ₃	7	1
CaO	65	39
MgO	1	1

Following heat data is given for both types of manufacturing process.

Item	kCal/kg of Ordinary Clinker	kCal/kg of Low Heat Clinker
Heat output excluding heat of formation	$360+6.7m_{\text{fuel}}$	$400+7.1m_{\text{fuel}}$
Total Heat Input	$27+6000m_{\text{fuel}}$	$27+6000m_{\text{fuel}}$

Calculate the following:

- Heat of formation of clinker of both the types. 8 marks
- Total heat output in terms of m_{fuel} 2 marks
- Amount of fuel needed per kg of clinker for each type of clinker from heat balance principle. ...4 marks
- Which type of clinker consumes more energy? 1 mark
- If cost of coal is ₹ 6,000/- per ton, what is the difference between energy consumption cost per ton of clinker, in Ordinary Clinker and Low Heat Clinker? 5 marks

Ans

Heat of formation of clinker

$$\Delta H_R = 2.22 \text{ Al}_2\text{O}_3 + 6.48 \text{ MgO} + 7.646 \text{ CaO} - 5.116 \text{ SiO}_2 - 0.59 \text{ Fe}_2\text{O}_3.$$

Constituent	Percentage of constituent in		Multiplier	Heat in kcal/kg of Clinker	
	Ordinary Clinker	Low Heat Clinker		Ordinary Clinker	Low Heat Clinker
SiO ₂	20	44	-5.116	-20x5.116= -102.32	-44x5.116= -225.10
Fe ₂ O ₃	7	15	-0.59	-7x0.59= -4.13	-15x0.59= -8.85
Al ₂ O ₃	7	1	2.22	7x2.22= 15.54	1x2.22= 2.22
CaO	65	39	7.646	65x7.646= 496.99	39x7.646= 298.19
MgO	1	1	6.48	1x6.48= 6.48	1x6.48= 6.48
Heat of formation of clinker				412.56	72.94
Additional Heat Output				360+ 6.7m _{fuel}	400+ 7.1m _{fuel}
Total Heat Output				772.56+ 6.7m _{fuel}	472.94+ 7.1m _{fuel}
Total Heat Input				27+ 6000m _{fuel}	27+ 6000m _{fuel}
Balancing the Heat Output with Heat Input				772.56+6.7m _{fuel} =27+ 6000m _{fuel}	472.94+7.1m _{fuel} =27+ 6000m _{fuel}
Mass of fuel consumed				0.1244 kg/kg Clinker	0.0744 kg/kg Clinker
Fuel Consumed per ton of clinker				0.1244 ton/ton Clinker	0.0744 ton/ton Clinker
Fuel cost				0.1244 x 6000= 746.4 ₹/ton	0.0744 x 6000= 446.4 ₹/ton

Therefore, the Ordinary Clinker consumes more energy.

The difference between energy consumption cost of Ordinary clinker and the Low Heat Clinker is

$$= 746.4 - 446.4$$

$$= 300 \text{ ₹/ton.}$$

Or

N4 C	A) In a textile mill 150 kg of fabric is dyed per batch in a jigger. The dye liquor is heated from 25° C to 90° C. The specific steam consumption is 0.65 kg/kg (with enthalpy 660 kCal/kg). Calculate the liquor ratio allowing 15% margin for the losses. 5 marks B) In a textile mill a thermic fluid heater of 20 lakh kCal/hour is meeting process heat requirement. Outlet temperature of fluid : 280°C Return temperature of fluid : 260°C Specific Heat of fluid : 0.55 kcal/kg °C Density of fluid : 840 kg/m³ Current coal consumption : 400 kg/hour GCV of coal : 4000 kcal/kg % loading of the thermic fluid heater : 45% 1) What is the thermic fluid circulation rate? ...3 marks 2) What is the existing thermal efficiency? ...2 marks 3) By providing an air pre-heater the efficiency of the thermic fluid heater improved to 62%. Calculate the coal savings for 6000 Hrs of annual operation? ...5 marks C) The operating details of a stenter in a textile unit are given below: - The inlet and outlet conditions of the cloth are shown in the figure below. The feed rate of wet cloth to the stenter is 1000 kgs/hour. - The heat input to the stenter is provided by a thermic fluid heater fired by firewood as fuel. - Gross Calorific Value (GCV) of Firewood is 3500 kCal/kg. - Firewood Consumption is 300 kg/hr. - The efficiency of the thermic fluid heater is 65%.  <pre> graph LR In[In 30degC 60% Moisture] --> Stenter[Stenter] Stenter --> Out[Out 5% Moisture 80degC] </pre> Calculate the drying efficiency of the stenter. (5 Marks)
Ans	A) Steam quantity needed per batch = quantity of fabric x specific steam consumption = 150 kg x 0.65 kg/kg = 97.5 kg Heat energy consumed per batch = Quantity of steam x enthalpy of steam = 97.5 kg x 660 kcal/kg = 64350 kCal/batch Heat energy consumed per batch = textile mass x liq. Ratio x 1.15(incl margin) x ΔT

	$64350 = 150 \times L \times 1.15 \times (90-25)$ $\therefore \text{Liquor Ratio} = 5.74$ Therefore liquor ratio of the jigger is 1:5.74
B)	
1)	Capacity of thermic fluid heater = 20 lakh kcal/hour Heat Duty = Capacity x % loading $= 2000000 \times 0.45$ $= 900000 \text{ kcal/hr.}$ Heat Duty = Circulation rate x density x sp. Heat x ΔT $900000 = Q \times 840 \times 0.55 \times (280-260)$ $\therefore \text{Thermic fluid Circulation rate } Q = 97.40 \text{ m}^3/\text{hour}$
2)	Input Energy = Coal consumption x GCV $= 400 \times 4000$ $= 1600000 \text{ or } 1.6 \times 10^6 \text{ kcal/hour}$ Efficiency of thermic fluid heater = $900000/1600000$ $= 0.5625 \text{ i.e. } 56.25\%$
3)	Efficiency of thermic fluid heater: 62% New coal consumption for thermic fluid heater = $0.5625 \times 400/0.62 = 363 \text{ kg/hr}$ Annual Coal Savings = $(400-363) \times 6000/1000 = 222 \text{ T/Hr.}$
C)	
Calculation of Stenter Efficiency	
1.	1000 kg/hr wet cloth contains 60% moisture (water) = $1000 \times 0.6 = 600 \text{ kg/hr moisture.}$
2.	1000 kg/Hr wet cloth contains 40% Bone dry cloth = $1000 \times 0.4 = 400 \text{ kg/hr Bone dry cloth.}$
3.	Weight of wet cloth at outlet contains 5% moisture = $400/0.95 = 421 \text{ kg/hr}$
4.	Quantity of Moisture at the outlet = $421 - 400 = 21 \text{ kg/hr.}$
5.	Moisture evaporated = $(600 - 21) = 579 \text{ kg/hr}$
6.	Heat energy required to evaporate 579kg/hr of water = $m \times c_p \times \Delta T + m \times 540$ $= 579 \times 1 \times (80 - 30) + (579 \times 540)$ $= (579 \times 50) + (579 \times 540)$ $= 341610 \text{ kCal/hr}$
7.	Heat Energy provided by firewood heated thermic fluid at 65% efficiency $= 300 \text{ kg/hr} \times 3500 \text{ kCal/kg} \times 0.65$ $= 682500 \text{ kCal/hr}$
8.	Stenter Efficiency = $341610/682500 = 50\%$
	Or
N4 D	A) Determine the cooling load of a commercial building based on the following data Outdoor Conditions

- DBT 36°C
- WBT 26°C
- Humidity 19 gm of water/kg of dry air

Desired Indoor Conditions

- DBT 25°C
- RH 50%
- Humidity 11 gm of water/kg of dry air
- Total area of wall 45 m²
- Total area of window 15 m²
- U - Factor (Wall) 0.35 W/m²K
- U - Factor (Roof) 0.363 W/m²K
- U - Factor (Windows) 3.00 W/m²K

Other Data

- Roof Area 20 m x 20 m
- CLTD
 - Wall 10°C
 - Roof 40°C
 - Window 8°C
- SCL Window 500 W/m²
- Shading coefficient (Window) 0.84
- Space occupancy 30 people
- Heat Gain/person
 - Sensible 80 W
 - Latent 50 W
- CLF for people 0.9
- Fluorescent light in space 21 W/m²
- CLF for lighting 0.9
- Ballast factor 1.2
- Computers and office equipment heat production 6.3 W/m² of sensible heat
- Air changes/hr of infiltration 0.25
- Height of building 3.9 m
- Product of density and specific heat of air 1210 J/m³°K
- Latent heat factor 3010 J kg dry air/m³ grams of water ...14 marks

B) The management has decided to give additional insulation to the roof at the cost of ₹40,000/-. This treatment modifies the U factor of the roof to value 0.300 W/m²K. The other data is

- The COP of the vapour compression system 3.75
- The efficiency of the motor coupled with compressor 90%.

	• The number of hours of operation in a year for the building 6000 hours • The power cost is ₹ 10 /kWh Find out the simple payback period for this additional insulation.
Ans	Solⁿ: A) External heat gain a) Conduction heat gain through the wall = U-factor x net area of wall x CLTD = 0.35 x (45-15) x 10 = 105 W b) Conduction heat gain through the roof = U-factor x area of roof x CLTD = 0.363 x (20 x 20) x 40 = 5808 c) Conduction heat gain through the windows = U-fact. x net area of window xCLTD = 3 x 15 x 8 = 360 W d) Solar radiation through glass = surface area x shading coeff. xSCL = 15 x 0.84 x 500 = 6300 W 1. Internal heat gain a) Heat gain from people = sensible heat gain + latent heat gain Sensible heat gain = No. of people x sens. heat gain/person x CLF = 30 x 80 x 0.9 = 2160W Latent heat gain = No. of people x latent heat gain/person = 30 x 50 = 1500 W ∴ Heat gain from people = 2160 + 1500 = 3660 W b) Heat gain from lighting = (Energy Input x ballast factor x CLF) Energy Input = (Amount of lighting in space/ unit area) x floor area = 21 x (20 x 20) = 8400 W ∴ Heat gain from lighting = 8400 x 1.2 x 0.9 = 9072 W c) Heat generated by equipment 1. Sensible heat gain by computers and office equipment = 6.3 x 20 x 20 = 2520 W Total heat generated by equipment = 2520 d) Heat gain by air infiltration = (Sensible + latent) heat gain Sensible heat gain = 1210 x airflow x ΔT Airflow = (Vol. of space x air change rate)/3600 = {(20 x 20 x 3.9) x 0.25}/3600 = 0.11 m³/s ∴ Sensible heat gain = 1210 x 0.11 x (36-25) = 1464.1W

$$\begin{aligned}
 \text{Latent heat gain} &= 3010 \times \text{airflow} \times \Delta h \\
 &= 3010 \times 0.11 \times (19-11) \\
 &= 2649 \text{ W}
 \end{aligned}$$

No.	Space Load Component	Heat Load	
		Sensile	Latent
1	Conduction through exterior wall	105	
2	Conduction through roof	5808	
3	Conduction through windows	360	
4	Solar radiation through windows	6300	
5	Heat gain from people	2160	1500
6	Heat gain from lighting	9072	
7	Heat gain from equipment	2520	
8	Heat gain by air infiltration	1464.1	2649
Total Space Cooling Load		27789.1	4149
Total Cooling Load = 31938.1 W			

B) Modified conduction heat gain through the roof $= U\text{-factor} \times \text{net area of roof} \times \text{CLTD}$
 $= 0.3 \times (20 \times 20) \times 40 = 4800 \text{ W}$

Savings in cooling load $= 5808 - 4800 = 1008 \text{ W}$

Savings in TR rating $= \text{Total savings in cooling load} / 3516$
 $= 1008/3516 = 0.287 \text{ TR}$

$\text{COP} = (\text{Refrigeration effect kCal/hr})/(\text{Power input kCal/hr})$

Savings in Power input $= \text{Refrigeration effect} / \text{COP}$
 $= 0.287 \times 3024/(3.75 \times 860)$
 $= 0.270 \text{ kW}$

Annual energy savings $= \text{Energy savings} \times \text{hours}$
 $= 0.270 \times 6000$
 $= 1620 \text{ kWh}$

Annual cost savings $= \text{Energy savings} \times \text{Energy rate}$

	$= 1620 \times 10$ $= 16200 \text{ ₹/year}$ Simple payback period $= ₹40,000 / ₹16200$ $= 2.47 \text{ years}$
--	--

..... **End of Section - III**